

A Catalog of Ghosts: the 120 Pilot Candidate Counterexamples and Sturmian Realizability Data

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Abstract

This catalog accompanies the Shadow Frontier project (successor to *Finite Spectral Shadows for the Collatz Valuation Cocycle*). It lists the 120 candidate-counterexample valuation words of pilot run 001 (deterministic regeneration, seed 1): words of length $M = 240$ over the alphabet $\{1, 2, 4\}$, calibrated to mean $\log_2 3$ within $2/M$ and carrying a non-coboundary order-3 shadow bias, sampled from the law $p_1 = 0.567$, $p_2 = 0.357$, $p_4 = 0.076$. Every one of the 120 is a *mixed-adic ghost*: its 2-adic realizability class modulo 2^{B_M+1} has least positive representative of nearly full bit length, not stabilizing between prefix lengths $M/2$ and M — the signature that no positive integer realizes the word. We also list ten Sturmian ghosts of the critical slope $\alpha = 2 - \log_2 3$ (all Sturmian words are now *provably* non-realizable, by the repetition-rigidity theorem); their data is shown for comparison. The objects below are points of $\mathbb{Z}_2$; the Collatz conjecture asserts that classes like these never meet $\mathbb{Z}^+$.

1 Conventions

For a valuation word $w = (b_0, \dots, b_{M-1})$, $B_N = \sum_{n < N} b_n$; the set of odd x whose first N valuations equal the prefix is a single residue class modulo 2^{B_N+1} ; “rep bits” is the bit length of its least positive representative r_N , “mod bits” is $B_N + 1$, and the fill ratio is their quotient. A positive realization would force r_N to stabilize at $\lceil \log_2 x_0 \rceil$ once $2^{B_N} > x_0$; a ghost rides the diagonal $r_N \sim 2^{B_N}$. The bias is $|D_3| = \left| \frac{1}{M} \sum_n \zeta_3^{-b_n} - \mathbb{E}_{\text{geom}}[\zeta_3^{-b}] \right|$, the order-3 non-coboundary two-point shadow.

2 The 120 pilot candidates: statistics

All 120 are ghosts: no representative stabilized; fill ratios ≈ 1 .

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
1	1.5833	0.214	225	380	381	381	1.000
2	1.5792	0.214	225	379	379	380	0.997
3	1.5792	0.214	225	379	378	380	0.995
4	1.5833	0.239	205	380	381	381	1.000
5	1.5917	0.220	213	382	383	383	1.000
6	1.5792	0.215	225	379	379	380	0.997
7	1.5917	0.215	225	382	383	383	1.000
8	1.5792	0.216	225	379	380	380	1.000
9	1.5792	0.229	214	379	380	380	1.000
10	1.5792	0.233	225	379	380	380	1.000
11	1.5792	0.229	225	379	379	380	0.997
12	1.5792	0.216	225	379	380	380	1.000
13	1.5917	0.233	193	382	382	383	0.997

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
14	1.5792	0.215	224	379	380	380	1.000
15	1.5917	0.216	208	382	383	383	1.000
16	1.5833	0.214	225	380	378	381	0.992
17	1.5917	0.215	225	382	380	383	0.992
18	1.5792	0.218	225	379	380	380	1.000
19	1.5917	0.220	225	382	383	383	1.000
20	1.5792	0.260	214	379	380	380	1.000
21	1.5792	0.272	208	379	380	380	1.000
22	1.5917	0.220	214	382	383	383	1.000
23	1.5875	0.228	225	381	382	382	1.000
24	1.5875	0.246	214	381	380	382	0.995
25	1.5917	0.222	211	382	383	383	1.000
26	1.5917	0.242	189	382	383	383	1.000
27	1.5792	0.222	222	379	376	380	0.989
28	1.5917	0.214	211	382	383	383	1.000
29	1.5875	0.215	225	381	382	382	1.000
30	1.5792	0.260	217	379	375	380	0.987
31	1.5917	0.214	216	382	382	383	0.997
32	1.5833	0.239	203	380	380	381	0.997
33	1.5792	0.218	225	379	377	380	0.992
34	1.5875	0.217	225	381	380	382	0.995
35	1.5917	0.214	224	382	379	383	0.990
36	1.5917	0.216	225	382	377	383	0.984
37	1.5792	0.220	225	379	380	380	1.000
38	1.5917	0.225	225	382	383	383	1.000
39	1.5875	0.221	225	381	381	382	0.997
40	1.5792	0.218	225	379	379	380	0.997
41	1.5917	0.225	206	382	380	383	0.992
42	1.5792	0.238	212	379	377	380	0.992
43	1.5792	0.215	225	379	375	380	0.987
44	1.5917	0.215	225	382	383	383	1.000
45	1.5875	0.217	225	381	379	382	0.992
46	1.5792	0.215	219	379	378	380	0.995
47	1.5792	0.229	223	379	380	380	1.000
48	1.5792	0.215	225	379	377	380	0.992
49	1.5917	0.215	225	382	383	383	1.000
50	1.5917	0.216	225	382	382	383	0.997
51	1.5792	0.229	223	379	377	380	0.992
52	1.5792	0.248	206	379	380	380	1.000
53	1.5917	0.218	224	382	376	383	0.982
54	1.5792	0.215	225	379	380	380	1.000
55	1.5917	0.233	225	382	382	383	0.997
56	1.5917	0.279	194	382	383	383	1.000
57	1.5792	0.215	225	379	380	380	1.000
58	1.5792	0.218	225	379	377	380	0.992
59	1.5792	0.238	201	379	380	380	1.000
60	1.5792	0.214	225	379	380	380	1.000
61	1.5792	0.238	225	379	380	380	1.000
62	1.5833	0.230	213	380	381	381	1.000
63	1.5792	0.222	219	379	379	380	0.997
64	1.5792	0.215	219	379	379	380	0.997
65	1.5792	0.222	225	379	380	380	1.000
66	1.5917	0.216	196	382	383	383	1.000
67	1.5917	0.215	225	382	383	383	1.000

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
68	1.5792	0.216	225	379	372	380	0.979
69	1.5917	0.216	225	382	382	383	0.997
70	1.5917	0.216	225	382	383	383	1.000
71	1.5917	0.214	224	382	382	383	0.997
72	1.5792	0.214	225	379	380	380	1.000
73	1.5875	0.221	225	381	379	382	0.992
74	1.5917	0.215	222	382	383	383	1.000
75	1.5792	0.215	225	379	380	380	1.000
76	1.5792	0.225	225	379	380	380	1.000
77	1.5833	0.241	225	380	381	381	1.000
78	1.5917	0.215	225	382	382	383	0.997
79	1.5875	0.221	225	381	380	382	0.995
80	1.5875	0.217	225	381	381	382	0.997
81	1.5917	0.225	220	382	380	383	0.992
82	1.5833	0.214	222	380	381	381	1.000
83	1.5917	0.215	225	382	381	383	0.995
84	1.5917	0.218	225	382	383	383	1.000
85	1.5792	0.222	225	379	378	380	0.995
86	1.5792	0.238	225	379	380	380	1.000
87	1.5792	0.229	225	379	376	380	0.989
88	1.5875	0.227	225	381	382	382	1.000
89	1.5792	0.218	225	379	380	380	1.000
90	1.5917	0.216	225	382	383	383	1.000
91	1.5792	0.216	225	379	378	380	0.995
92	1.5833	0.225	225	380	381	381	1.000
93	1.5917	0.220	220	382	382	383	0.997
94	1.5792	0.214	225	379	380	380	1.000
95	1.5792	0.238	223	379	380	380	1.000
96	1.5792	0.238	225	379	379	380	0.997
97	1.5917	0.233	221	382	383	383	1.000
98	1.5792	0.229	219	379	380	380	1.000
99	1.5917	0.218	223	382	382	383	0.997
100	1.5792	0.215	225	379	379	380	0.997
101	1.5792	0.218	225	379	380	380	1.000
102	1.5792	0.218	216	379	376	380	0.989
103	1.5792	0.229	225	379	379	380	0.997
104	1.5875	0.217	225	381	382	382	1.000
105	1.5917	0.229	211	382	383	383	1.000
106	1.5833	0.232	225	380	380	381	0.997
107	1.5792	0.218	225	379	379	380	0.997
108	1.5792	0.214	225	379	380	380	1.000
109	1.5917	0.220	225	382	381	383	0.995
110	1.5917	0.220	225	382	383	383	1.000
111	1.5792	0.229	225	379	380	380	1.000
112	1.5833	0.218	225	380	379	381	0.995
113	1.5792	0.248	225	379	379	380	0.997
114	1.5792	0.229	223	379	380	380	1.000
115	1.5917	0.220	224	382	383	383	1.000
116	1.5917	0.233	200	382	382	383	0.997
117	1.5792	0.222	225	379	378	380	0.995
118	1.5875	0.214	225	381	382	382	1.000
119	1.5792	0.218	224	379	380	380	1.000
120	1.5792	0.214	225	379	380	380	1.000

3 Sturmian ghosts (critical slope, now a theorem)

Words $b_n = 1 + \mathbf{1}\{\{n\alpha + \theta\} \geq \alpha\}$, $\alpha = 2 - \log_2 3$. By the repetition-rigidity theorem these are non-realizable for *every* slope and intercept; the data below shows the same diagonal-riding signature as the candidates.

θ	$ D_3 $	B_{240}	rep bits	mod bits	fill	rep bits at $N=100/200$
0.000	0.447	380	381	381	1.000	159/316
0.100	0.447	380	380	381	0.997	157/318
0.200	0.447	380	377	381	0.990	159/317
0.330	0.447	380	381	381	1.000	155/318
0.500	0.453	381	379	382	0.992	159/317
0.610	0.453	381	376	382	0.984	160/318
0.750	0.453	381	382	382	1.000	160/317
0.900	0.447	380	381	381	1.000	160/318
0.057	0.447	380	381	381	1.000	159/318
0.085	0.447	380	381	381	1.000	159/318

Example Sturmian words (first 120 letters):

$\theta = 0.00$:

12212122121221212121212212121221212122121221212212121212
21212212121221212212121221212212122121221212212122121221

$\theta = 0.10$:

122121212212122121221212122121221212212122121212212121212
1221212212122121212212122121221212212122121221212212121212

$\theta = 0.20$:

121221212212122121221212212122121212212121221212122121212
1221212122121221212212122121221212212122121221212212121212

$\theta = 0.33$:

1212212121221212212121221212212122121221212212121221212212
1212212122121221212212122121221212212122121221212212121212

4 The negative cycles, computed explicitly

The three known negative cycles anchor the ghost families: each periodic valuation word below has its unique 2-adic realization at a *negative* integer, so the least positive representatives of its realizability classes ride the ceiling $2^{B_{N+1}} - |x|$ — the periodic ghost lines of the trajectory zoo. Throughout, $S(x) = (3x + 1)/2^{v_2(3x+1)}$ and a period- p word w with sum B satisfies the affine cycle equation

$$x = \frac{A_w}{2^B - 3^p}, \quad A_w = \sum_{j=0}^{p-1} 3^{p-1-j} 2^{B_j}, \quad B_j = b_1 + \dots + b_j.$$

Negative cycles are exactly those with $2^B < 3^p$ (denominator negative).

$x = -1$: **word (1), period 1**

$$3(-1) + 1 = -2 = -2^1 \cdot 1, \quad v_2 = 1, \quad S(-1) = \frac{-2}{2} = -1.$$

Cycle formula: $p = 1$, $B = 1$, $A_w = 3^0 2^0 = 1$, and $x = 1/(2^1 - 3^1) = 1/(-1) = -1$. This is the 2-adic limit of the all-ones word: $b \equiv 1$ forces $x \equiv -1 \pmod{2^k}$ for every k .

$x = -5$: **word** $(1, 2)$, **period** 2

$$\begin{aligned} 3(-5) + 1 &= -14 = -2^1 \cdot 7, & v_2 &= 1, & S(-5) &= -7, \\ 3(-7) + 1 &= -20 = -2^2 \cdot 5, & v_2 &= 2, & S(-7) &= -5. \end{aligned}$$

Cycle formula: $p = 2$, $B = 1 + 2 = 3$, partial sums $B_0 = 0$, $B_1 = 1$:

$$A_w = 3^1 2^0 + 3^0 2^1 = 3 + 2 = 5, \quad x = \frac{5}{2^3 - 3^2} = \frac{5}{8 - 9} = -5.$$

Letter mean $\frac{3}{2} < \log_2 3$: in the positive world this word ascends, the gentle green dotted line of the trajectory zoo.

$x = -17$: **word** $(1, 1, 1, 2, 1, 1, 4)$, **period** 7

$$\begin{aligned} 3(-17) + 1 &= -50 = -2^1 \cdot 25, & S: -17 &\mapsto -25, \\ 3(-25) + 1 &= -74 = -2^1 \cdot 37, & -25 &\mapsto -37, \\ 3(-37) + 1 &= -110 = -2^1 \cdot 55, & -37 &\mapsto -55, \\ 3(-55) + 1 &= -164 = -2^2 \cdot 41, & -55 &\mapsto -41, \\ 3(-41) + 1 &= -122 = -2^1 \cdot 61, & -41 &\mapsto -61, \\ 3(-61) + 1 &= -182 = -2^1 \cdot 91, & -61 &\mapsto -91, \\ 3(-91) + 1 &= -272 = -2^4 \cdot 17, & -91 &\mapsto -17. \end{aligned}$$

Cycle formula: $p = 7$, $B = 1+1+1+2+1+1+4 = 11$, so $2^B = 2048 < 2187 = 3^p$; partial sums $(B_0, \dots, B_6) = (0, 1, 2, 3, 5, 6, 7)$:

$$\begin{aligned} A_w &= 3^6 \cdot 1 + 3^5 \cdot 2 + 3^4 \cdot 4 + 3^3 \cdot 8 + 3^2 \cdot 32 + 3 \cdot 64 + 128 = 729 + 486 + 324 + 216 + 288 + 192 + 128 = 2363, \\ x &= \frac{2363}{2^{11} - 3^7} = \frac{2363}{2048 - 2187} = \frac{2363}{-139} = -17 \quad (139 \cdot 17 = 2363). \end{aligned}$$

Letter mean $\frac{11}{7} \approx 1.571 < \log_2 3 \approx 1.585$: again sub-critical, again a positive-world ascender. Note how close $11/7$ is to $\log_2 3$ — the -17 cycle is a near-balanced word, the best rational approximant realizable at small height, and exactly the shape that the Baker cycle bound ($x_{\min} \leq Cp^{\tau+1}$) governs.

5 The June 2026 constraints, and the surviving candidate list

Since this catalog was first compiled, the program proved a stack of further theorems (companion notes: narrow-band exclusion; two clocks; universal ghost clock; CIC round chain; faithfulness). This section records what they jointly do to the counterexample profile, and then lists what now survives.

The new constraints, in catalog form

- *Exact narrow-band exclusion.* No acyclic orbit dwells more than $CY^{\theta(D)}$ consecutive steps in a band $[Y, 2^{2D}Y]$; the count of confined words is an exact integer (no Perron estimates), and the counting capacity is exhausted at $D^* = 1.246$ (band ratio 5.63). Bounded-discrepancy words ($\sup |S_j| < 1.246$) are unconditionally non-realizable by divergent orbits — the Sturmian section above is now a corollary. In a ratio- ≤ 2 band the dwell bound is *logarithmic* (the dyadic strip carries one word per phase).
- *The phase diagram has two arithmetic lines.* Vertical: $D^* = 1.246$. Horizontal: the binary bias ceiling $\varepsilon_{\text{bin}} = 0.449$ — calibrated $\{1, 2\}$ -words cannot carry more order-3 shadow bias than this; exceeding it requires letters $b \geq 3$, which are descent-heavy. Saturation $\varepsilon^* \approx 0.54$.

- *Every letter is a metered ghost approach.* $b=1$ runs meter -1 (length exactly $v_2(x+1)-1$); $b=2$ runs meter the trivial cycle $+1$ (depth decrement exactly 2, parity handoff); $b \geq 3$ letters read $v_2(x + \frac{1}{3})$. Universally: every signature σ of depth K has phantom root $\rho_\sigma = C_\sigma/(2^K - 3^\ell)$ (the cycles above are instances), alignment drops exactly K per ridden block, rides last exactly $\lfloor (a-1)/K \rfloor$ blocks, and alignment a at height x costs $2^a \leq |2^K - 3^\ell|x + C_\sigma$. Amortized over *all* ghost families of depth ≥ 3 , the available drift gain is $R = 0.0882$ bits per letter against a 0.415-bit budget.
- *Supply is consumed without replacement.* An acyclic orbit visits each integer once: each dyadic band holds only 2^{k-g} points of depth g per ghost, ever. Consequences: at most 2^{k-3} upward crossings of level 2^k ; a divergent orbit spends $O(X \log X)$ lifetime steps below X .
- *The survival coin.* Parsed at 1-run starts, life against the letters- ≤ 2 regime is the reading $v_2(3^\ell m - 1) = v_2(m - 3^{-\ell})$, death iff odd. On Haar fibers the coin is exactly i.i.d. fair-thirds; on ghost-aligned seeds it is deterministic (LTE): Mersenne $2^g - 1$ survives round one iff g is odd with $v_2(g-1)$ even; the 3-smooth families $2^g 3^a \mp 1$ cascade in closed form, with round-two readings such as $v_2(a' + 2 - \text{Log}_3(-5))$ — the 2-adic *logarithm of the -5 ghost above* — and every tested seed dies by round 7.
- *The $\{1,2\}$ alphabet has an entropy deficit.* $\theta_{12}(D) \leq 0.97907 < 1$ at every band width: for an orbit with no letter ≥ 3 the phase diagram's allowed region is empty — no band of any ratio offers refuge.
- *The blind spot bites.* A positive integer has $\log_2 x_0$ bits, period: after $\approx \frac{2}{3} \log_2 x_0$ steps the word has consumed them all and every later letter is forced by the $\times 3$ carry cascade. The infinite word has Kolmogorov complexity $\leq \log_2 x_0 + O(1)$ while repetition rigidity demands near-maximal factor complexity in banded windows. Realizable words are a countable, computable family (of the 2^{30} depth-30 words over $\{1,2\}$, exactly 66 are realized below 10^6).

The composite profile

A counterexample must therefore be: a divergent orbit, fleeing upward (escape $O(X \log X)$, single excursions below X sublinear), never pausing in any narrow band, calibrated to mean $\log_2 3$ yet carrying non-coboundary bias below the ceilings (0.449 binary, ≈ 0.54 saturated), financing every ascent through without-replacement ghost-depth pools and an amortized 0.088-bit census margin, beating an i.i.d. fair-thirds coin forever on its $+1$ -handoff parities, with a word of near-maximal local diversity generated — after a free opening of $\frac{2}{3} \log_2 x_0$ moves — by a fixed finite carry transducer with no further choices.

The candidate list, June 2026

No specific integer candidate exists: every concrete object ever computed in this program is either a 2-adic ghost or dies with a finite certificate. What survives is six structural profiles. For each we list its best known approximant and the meter that taxes it.

profile	best known approximants	taxing mechanism / status
C1. Carry-cascade pseudo-random divergent (generic profile: calibrated, biased, high-complexity)	the 120 pilot words of this catalog (all ghosts); integer side: survival champions 27, 4591, 31911, 517191, 687871 (depths 17, 27, 33, 38, 50; 59–76% of each life is in the forced carry regime)	fair-thirds survival coin; census margin $4.7\times$; all known members die; champion height grows $\approx 2^{D/2.2}$ with depth
C2. Binary violator: letters $\{1, 2\}$ forever ($\mathcal{C}_\infty \cap \mathbb{N}$, the CIC class)	none positive; the 2-adic members are the Sturmian-type ghosts of §3 and their high-complexity cousins	entropy deficit $\theta_{12} < 0.9791$: no band of any width is a refuge; parity coin must read even forever
C3. Parity-channel rider (word biased toward $(1, 1, 2)^\infty$)	integers 2-adically near $-19/11$; least representatives ride the full diagonal (ghosts)	the $K=4$ clock of $-19/11$: alignment a costs height $11x + 19 \geq 2^a$, rides last $\lfloor (a-1)/4 \rfloor$ blocks
C4. Ghost-aligned seeds (3-smooth neighborhoods $2^g 3^a \mp 1$, incl. Mersenne)	round-one survivors $2^5-1, 2^{13}-1, 2^{17}-1, \dots$ ($g-1 \equiv 2^{2t} \pmod{2^{2t+1}}$); family champion $2^{705}-1$ ($n=704$: six rounds, readings 8, 2, 2, 2, 2, 2, 3)	deterministic LTE cascade; ghost-log readings; every tested seed ($n \leq 3000$) certified dead by round 7
C5. High cycle (near-balanced periodic word, B/p a convergent of $\log_2 3$: $19/12, 65/41, 84/53, 485/306, \dots$)	the $-1, -5, -17$ cycles above (negative side); no positive candidate below current k -cycle exclusions	cycle formula $x = A_w/(2^B - 3^p) + \text{Baker pinning } x_{\min} \leq Cp^{\tau+1} + \text{narrow-band period localization}$
C6. Exponent-level conspirator (round-chain exponents tracking 2-adic digits of ghost logarithms, e.g. $N_0 = \text{Log}_3(-5) = 2556611305511861515 \pmod{2^{64}}$)	none known; alignment at exponent level costs height at exponent level	the lifted clock: ghosts-of-ghosts; p -adic Baker bounds the readings effectively

The list should be read as a specification of where a counterexample would have to hide, not as evidence that one exists: profiles C2–C6 are each pinned by an exact meter with a positivity cap, and C1 — the generic impostor — is the quenched calibration wall itself, the single open core to which every known framework (ours, Tao’s, the LLM-collaboration manuscript audited in the companion notes) independently reduces.

6 Collatz Elevator Pitch

One-liner. "We can prove the average number loses this game. We can’t prove that this particular number does — and every number is a particular number."

Thirty seconds. "The rule is simple: take a number — if it’s odd, triple it and add one; if it’s even, halve it. The conjecture says everyone eventually crashes down to 1. Now, the ups and downs of any number look exactly like coin flips, and the coin is rigged downward — so on average, everything falls. That part is basically understood. The problem is that nothing here is actually a coin flip. Each number’s entire future — every up, every down, forever — was fixed the moment you chose it, written into its digits. So to prove the conjecture, you’d have to rule out that somewhere out there is one number whose predetermined script just happens to dodge the crash forever. Probability can’t do that: probability talks about most numbers, never about that one. It’s the same wall as: we can prove almost every number has perfectly shuffled digits, but nobody can prove $\sqrt{2}$

does — even though $\sqrt{2}$ is sitting right there, fully determined."

The twist, if she wants one more floor. "Here's what makes it maddening — and what the last year of work actually showed. We've cornered the hypothetical escape artist. It can't follow any pattern: we proved that if its moves ever repeat, even once, in the wrong circumstances, that repetition snaps shut into a loop, and all loops are dead. So a survivor would have to move in a way that's indistinguishable from pure chance — never repeating, perfectly balanced on the knife edge between rising and falling. But it also can't actually be random, because random walkers provably crash. It would have to be a perfect impostor: a fully scripted number flawlessly impersonating chance, forever, while hiding a secret bias it needs to stay alive. We're fairly sure that impostor can't exist — being biased and looking unbiased pull against each other — but proving a specific, predetermined thing can't perfectly imitate randomness is something mathematics doesn't yet know how to do. That's the whole obstruction, in one sentence: we can beat every strategy, but we can't beat 'no strategy.'"

A note on your original phrasing: "deterministic numbers that behave totally randomly" invites the response "then they crash, like random things do —done!" The fix is to put the burden where it really sits: deterministic numbers that we can't prove aren't perfectly disguised as random. The casino version also lands well: "We've proven the house always wins on average. The conjecture asks us to prove no individual gambler beats the house forever — when every gambler's every move was decided before they walked in."

A The 120 candidate words

Alphabet $\{1, 2, 4\}$, length 240, row-wrapped at 60 letters.

#1 (mean 1.583, $|D_3|=0.214$):

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111411211121121111112114421114112222111122212112122212112221
112221224121212222121212112111112122122211411112211121
22211111114222222111111212141212112142211414212111141
2112114122221211111112222212141212122141121111114422241

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#2 (mean 1.579, $|D_3|=0.214$):

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22222222222222211111121121121421111211421111221222221114241211
11111241211121142211114111211112211221221211111121211111211
2122221121211111422142411211121411211144212111114121114221
2121111111111221211111212121212121211111121141242222212222211

```

#3 (mean 1.579, $|D_3|=0.214$):

```

2221121111121124212211121112111212122111211112121221144412
42222221112212121111112111111214121211122242221411212111
112411112122242111121112112221212211112411211224111212121
224122211411211112222421212421222121114411211211212111122

```

[illegible][illegible]

```
#6 (mean 1.579, |D3|=0.215):
222222224222222421121111141242121141122122222111112112241221
111121212111111111112211112222242111421221421111112422221
111112221111112112221411122212121124211111411121211222
1111121111211111221221421114122121141214111121212121212
```

```
#7 (mean 1.592, |D3|=0.215):
1211121112112121111224121112121411212114112411221242222211
22122111111221111122121222421411211122112111222111211112112
21242214111122141242111122221411214211211121212111112211111
11114121222242111111221212211121112124212121141122122214211
```

#8 (mean 1.579, $|D_3|=0.216$):

```

12211111111212121111141121222211112111121111211142211414
222221141211111421222112211111121141221212112212121421121
4121211241112111242211221111111414212112222441112111221
2141222121141121112122211121224111112112124121221221222221

```

```
#9 (mean 1.579,  $|D_3|=0.229$ ):
2222222222222422222222222222222222222422211122211221144111
121121122111221112211111121114212411112221112112111122241111
```


2412121211411111211121111212211112211121111412121214211111
111122112211111211111142242121121112211111111112112421

#10 (mean 1.579, $|D_3|=0.233$):
111112111222121222121211114111414222441214211212124111122211
111111111121121421111411122221211224211422141122144142111121
1421411121221111121222241111111111111122422121212211111111
111221422111212111111114221121111212111211121114122221

#11 (mean 1.579, $|D_3|=0.229$):
222222422222422222112211221241221221112211142211112122112
1111121112112121111211122112112112211122241121211241141
12121111421112112112211141212121121422211112112411211222121
2242212122121121114212111214122111111221111111212211111

#12 (mean 1.579, $|D_3|=0.216$):
42221121211412112224411122221112411221221112121112441124421
11222211111211121112212111221221211111111121242212112122
114122214121111112412411112114112121111141112221141211212
2111121112111111222421121114211111212212121111212211221

#13 (mean 1.592, $|D_3|=0.233$):
111222212111121
1221242141122111111221111222122141114421122211212442114111
121241421122224122111422121211222421122114421142422121221111
211124144212122112121222111211124111211121121114212122141

#14 (mean 1.579, $|D_3|=0.215$):
242222222422222221111111211221111111111111111221121224211
2111212122221121221214121121221121221211212111121142111121
2121112211141111211212441222112111221112212412411211222212
211211111111212121411411112124111224141124122121111211411

#15 (mean 1.592, $|D_3|=0.216$):
11124411122111211112112211221
12122112114124121421124122111111224121122222211111222211122
12221222214212121214222241121411121121112221411414111122111
21121122112124211211412421121122411421222221114112222211111

#16 (mean 1.583, $|D_3|=0.214$):
21221411111211112121111212211121211121122414224112111221122
241122142121211111241211244411221221111112221212422211214
2111214211111121221111242112212114121111122112211111121222
114121211411211111241222121221211112121212212121112111211122

#17 (mean 1.592, $|D_3|=0.215$):
11121121224212122211221222111112122421112112112222114111212
21111124212241211211212111221141122141121124221112221211411
4111114212112211121122111221112111111221221121221121114112
212221111111114111411211214111111222112124211121122242421222

#18 (mean 1.579, $|D_3|=0.218$):
22222224242211124111122122121141122122112121221122114241112
11122421122111212111411112121141111222121121212412111121222
12114121111222221121111122114124141122111221111212212221111
21221422112111112121211111222114211211122111121111122111

#19 (mean 1.592, $|D_3|=0.220$):
11241111121111212111241121211121211114122111112111112221
121212111212212112121114111111112141122122442121122112222112
2114142111411121122122441111121121111211222141241114121222
1111211221244112121111221212411121212124242212112211111111

#20 (mean 1.579, $|D_3|=0.260$):
222222422222222222222222222222212122111112211111221124112
21111211221122412111411112221121221211222111111412122212212
42111421212422121211211221111112112211112211122124221111211
12121111112122212111121111111111141121112411212221222121111

#21 (mean 1.579, $|D_3|=0.272$):
2222222222222222222222222222222121211112212211121122222121
11111122121121111222111121111211112221112122111141121414121
111211211121422211211212121122421212211112414212111121121121
121121112111111221212112211211222122212441221211111412122

#22 (mean 1.592, $|D_3|=0.220$):
11111111111111111111111111111211112242124211142124122111112241
221114214212211122221221414421121112221121141222111211112212
11411111211141121211112111122441121211122112122242111211114
121111121211212124242212121241212211122411111212122122211212

#23 (mean 1.587, $|D_3|=0.228$):
142222112411411141112111121122414422122111111211211222412
2221211112112212111112111221141121212141111121111221122122
1242212121111141214211211222111121211114121112421211112112
1224111111111111111122221111224244111241142141111122111

[illegible]

14121444211112412422412112121112142121411211122111111241121
2211111411121112111211111111221122114222111111121214112

#39 (mean 1.587, $|D_3|=0.221$):
2241112122211411111224122211141121111111411212111241211211
111211122221214111422114112211221122121112122121111211111
24141121112124212121114222211114121121121112121212111111
14422121111212412212211122211121121114221212111224442112211

#40 (mean 1.579, $|D_3|=0.218$):
2222222222114112211111121411121111211112111211122241122111
112121411111224111211112222122112211122112112212111412142212
2221221222212211211112111411222112121111214111221122211411
12111121121241141221221221212121141122144121111111242111212

#41 (mean 1.592, $|D_3|=0.225$):
111111111111111111111111111112211211111122141221412211
2111122211121412211221111222111111221112112142124142421211
42212111114111114111124111212121222111222122214122221221212
2421212241114214422121111222111212411114121224211112412212

#42 (mean 1.579, $|D_3|=0.238$):
222222222222222222222222222211221114112111111141124121111
1212411124111221121211111121112112111122211112112212112121
2111141211111221111221221212121112121112112121111121221111
21224111241112212221221111221112422211411212121224222212441

#43 (mean 1.579, $|D_3|=0.215$):
2224222422211212242112141211121112441121211112111124111222
1111122211122241221111111112121211421121111212112224214221
22111121211112212211212111121111212211112221221121111122214
2122412112111122121211112112114221111114211111212221412121

#44 (mean 1.592, $|D_3|=0.215$):
111111111111111112111221221122111142111221111224211121211411
4222111112211111222442111112211112122412111212221412211422211
112214211221111211112212241121211221221121122121112111121214
112112111121411112222114122221121212221224411442221222112112

#45 (mean 1.587, $|D_3|=0.217$):
1211212242122111414111111111212122411221111111121111211112111
1221122221121121222411124114122224124214121121121221421412
1111111222122121114221111212211121121141212111222222211412
111111122121212111112121411124111141121222112142211111141

#46 (mean 1.579, $|D_3|=0.215$):
2222422222222222222222121111121111211214221114411221111112
1241111111121211111211122221111212212112111122111224122114
2111214441111142122111222211112111112122121121112222211111
2141121412412141121142111121212111211421121111211112121212

#47 (mean 1.579, $|D_3|=0.229$):
2242222222222222222222224422222212211112411114411421111111
1111212112111112414121111212212121112221121211211222111211
114111212111221111122121222211221121212111111112122211111
1214211211222122421112222122212122111111111112111212121411

#48 (mean 1.579, $|D_3|=0.215$):
242222224212211121111111121211124411411111122422121421112
211121222212221241112112212212212121212121242111141111211
1214141111111112111411211212222142121112122112122212111221
2424212212111111111112222111222112111411211111211121212112

#49 (mean 1.592, $|D_3|=0.215$):
111111111141121142222122114411211221421214242121121244211111
21222112112112121214212211411212111212212121222221212111211
221111121122111111121121241112111221111221212111212111122
2222111121141212222114421421222111112411112222412121111111

#50 (mean 1.592, $|D_3|=0.216$):
122112121421111242121122112121211221212221421421111121111121
1412211111414211112221221214121221111144112111111111121221
211144121112111211212122112211222212141212111111121114111212
121411242124212212122122124112211114121112112122421111221211

#51 (mean 1.579, $|D_3|=0.229$):
2222222221221242111224122211221121112111221122214122111112112
21112441121112142111211141111221111222442111111121122211121
212122121111122222111221211412212221121221112111121112212221
111112122221141114114112121111121411222121121221122111212112

#52 (mean 1.579, $|D_3|=0.248$):
2224222222222222222222222222221121111211111121111112121
21112221111111211124111112222112111111222111121121122111211
242111111122112122421222112442112414121111221112211121111111
41111221221112121111124112211211212114411211121121121221121

[illegible]

21411121411122122111214111211112121121221224121411122211121
412111144211212111212412211212122212111212122211222111211222

#68 (mean 1.579, $|D_3|=0.216$):
22222222222422424211121211112124122114421211211411111221
212112121111112442121221222211212111211121211211141211112
112121114122121111412111211111122122112211111112112411212
12212111114111111441122111111122421211212411122211112111

#69 (mean 1.592, $|D_3|=0.216$):
111111212241111221211114111112212112122411222122221142111
11114221211242212141122224242211214242111221111221211222111
1411111212121221122111121221112211411111242212111211111
12124112221121411111411122421121112111412121121111421212

#70 (mean 1.592, $|D_3|=0.216$):
211142121122412224111224112112112121121111412112222
112211121214121221212111111224112121121212111122111214224
1122211212112121211422112221121121141114222111222411111111
124122121114121111221212211211121442111114422412111111112

#71 (mean 1.592, $|D_3|=0.214$):
11111111111111112211214121421112122112211114122144221122222
1144144124242112221411121121212111112111421222112221212211
12222421122121112122212411111411211221121112121111212111222
2112122112142121211112111221121241211211221412121112111112

#72 (mean 1.579, $|D_3|=0.214$):
24224222221221111141112111241121121212211221111221212121
2244111142211221112241111121212111211212114212121141242111
2121112112221122214122411112111121221422121211212111122121
122212212121211211112111214112111121112241111111211411114

#73 (mean 1.587, $|D_3|=0.221$):
1211411111224111211212422112121221141122211112111124214211
1222211111112212111212141211144111221211112121111221212121
112211111112412421211222122111114121124111111121112112121241
21112224221411112241121121121222141212122241111221111411241

#74 (mean 1.592, $|D_3|=0.215$):
111111111111111112122112121422222114111122121212121122111
11122122114124211222211112212212121121222412112222142212212
212221221112121111122242122111111121212121211112114244142
2214112111241112114111211111122111141224221222211411211124

#75 (mean 1.579, $|D_3|=0.215$):
221124111421212121211122221111112421211212121121141221211
2121114121141111212111121141121241211241111211221221224412
2211112222111121221221111121212122221141112111422221111211
1142112211211212122121112122124124112121112122112111114121

#76 (mean 1.579, $|D_3|=0.225$):
2222242111221411122111112421111112111211212411111214111112
4211111211112121211112241211121421212111111112221122122
2111221212212111411211124221242111241122112111111112112414
2111412121121111211112122212141422111211221114412214421112

#77 (mean 1.583, $|D_3|=0.241$):
11111112222111211111211112221211414122242142222221242142
22112211412422121212211211124121112121112112111121222221
1211221121211422221114222211124111121121112211111221221122
1212212211242121112121122122121112212211122121111221212112

#78 (mean 1.592, $|D_3|=0.215$):
11111111111121112122211442211222212222211112112212111412
11222112221211424141111222121111211141112124411112112222111
2212221211212111211222144122111111141212222111121222212222
111122412111221112111211211142212221421411122111114411211

#79 (mean 1.587, $|D_3|=0.221$):
14121122111111421221212111212112222111211111114111111124
112114142221221111211221421222212111211221111114414124222
1111112411214111211122111112421141211114211221211122112112
12411214111222111221212211211241141112121221221121111222242

#80 (mean 1.587, $|D_3|=0.217$):
111112144211112211141212112111222111122121211121221111121
2112211111214142122121114221422121112121112122112121212142
121112212212122211412222121211121111122124111114211122242
12211224222111241114212121211211242421212111221112111121112

#81 (mean 1.592, $|D_3|=0.225$):
1111111111111111124141112222112141212121111412114211421
1122121111112112121212111122424111111221142112111212141
22221121112114221112224111242221111114122112122124122211112
1111121211122212112241142111242212421222142111112141221121

#82 (mean 1.583, $|D_3|=0.214$):
21211212424121111214112121121221121121211212422111111121214
224111211121124412221122122111222112121122141142214112424421
1242112211121111121122241111212121211122111211112112212121
1111111122122111214112211121111222111122112241211221221111

#83 (mean 1.592, $|D_3|=0.215$):
112212111124212112211211112112111141122121212221111111122
22111122411122221222122212111121222112112111422421441221212
241112221221211121414122442221112241111211212112211112111421
411112121212114141111122121221411211212111112221112121111

#84 (mean 1.592, $|D_3|=0.218$):
122111111121141121122212142221142212112221111212122121211211
121211112112122421411111141121242211112122221111211144112242
121111111122121122122221221144122111112122212211411111214
21112121422114211121211121112114112111122212112222221112221

#85 (mean 1.579, $|D_3|=0.222$):
2121224411222214222122121211112111121114141141211111122
1221211111211212212121121121221211412221211211121122221211
22212421221121112112112412121411112212411112211221211112212
44122122121211112212122411121212121222111211412121211111

#86 (mean 1.579, $|D_3|=0.238$):
222222222422242222222222211221111212241212211121242112214
112111111111112112214211111221112111112121121412221111142
211221111112211112111212111214122121212111122121221122112
2121222211141142421221121211211112222212121121114111112112

#87 (mean 1.579, $|D_3|=0.229$):
4222222222222224222422214222211122212121112111122111412
11122111211111422111221111121124112111221224111111121111
1121222111111111111114111224112122121111112211211242222
2211222211111211121124142112122112212224221111212121211221

#88 (mean 1.587, $|D_3|=0.227$):
2421412121212121214121114111222121211111212211212121212222
121112221111211111212412111211122122111111121111211212121
12124212241221211121212221111122121222211212421212121221122
11224211112222121244222111121414121111121142211111124122

#89 (mean 1.579, $|D_3|=0.218$):
42222222221141112112211121212122111111212111111122122121
121122211212122122211124121121114112114211121121211121114
241221111122111411112241411212111211112422212222221212242
211221111211212122111111222112111122411142141112412221112

#90 (mean 1.592, $|D_3|=0.216$):
1111111212121114212212121112122112211121211112142211142111
21112211112441211122221421211111112121121121142211112421112
111121212212222121224142112211121212121212111211411421211
1111211411221212111122221221411411112244142112111241224111

#91 (mean 1.579, $|D_3|=0.216$):
42222422221214111212112121214111211114111212111111211112
1121121212212411211121412212121111111112112224111211122212
1244122141421144121211211111241221112111122221221212111211
221221111124122111111121112211211112212114121114214121222

#92 (mean 1.583, $|D_3|=0.225$):
1111112122211122122114112112211224212222111124221211112112
124111222411211241121122112111111121224122422111111112212
2111211122121211121111221211121212242222222211122121212
11211211222124124242114121111211111221111121242111241211221

#93 (mean 1.592, $|D_3|=0.220$):
111111111111111111122124212111114141212221211112112212111
1122141111212211111122212412112111124121121212212122112122
44242112211142112411212111221121112112114111244114111124122
22111211121122212212111424212242121411212111411122112122111

#94 (mean 1.579, $|D_3|=0.214$):
4422222222222242412111111111111211122111121124142112211211
112112122121224212122411211112211122124212112111411121111211
1412421121111121221111224114121111124122111111111142111112
22211122111112121112142111212221221112112111121412212112211

#95 (mean 1.579, $|D_3|=0.238$):
2222222222222224222222222222222222221121111111112212121
241111112111112112122111241121112111112142121121121211121221
111241222211114111221221111142224121121111212112111111112
2111122122211141112212221111242224111111112111222412211111

#96 (mean 1.579, $|D_3|=0.238$):
222222242222222222212112242111121111212112124121211122122
22112222121111211141221111111121411221112211212122111221112

12111412211411112211121221111222112124141211212124221211141
122212111111121124122111112221212111111221121112421122121

#97 (mean 1.592, $|D_3|=0.233$):
11111111111111111211414222111121221112222111112111142412
211211421421114212121224111422124111212421114221122212211112
1122112111111121212122211121114121212222141214111121221
121411112221411111222112211141121141211241121224411112411

#98 (mean 1.579, $|D_3|=0.229$):
22222222222222222211141211111221112212221111112121221
1212111422112111221111412221221111221112111121112121111122
111212241411111141222221211112111121211212222114111124
124121222411211211112422412212114211211122111224112111212212

#99 (mean 1.592, $|D_3|=0.218$):
11111111111111114112211414242242111124124111122112112212
1221114141121112211112412212114112111122112124422221224221
12221212221212111111211222224214122211221112221122111212212
1221112111122221212221112211121112222141111121121212211

#100 (mean 1.579, $|D_3|=0.215$):
2222222111441121124221121114121221122212141112121111121221
2141122121222122212221222111212142112212211124411411211111
114411114112111411411111421122112211211211121212121212211
2121211111121212212111112221122212211121112112121411211

#101 (mean 1.579, $|D_3|=0.218$):
222222222222111121212111121121114212112121111221141241211
1211141111121121221111112121241122221122211212111211122111
14222112112112111411211121212421222211142422212211421221
2224112111412141212111121111211121212112112111424122111

#102 (mean 1.579, $|D_3|=0.218$):
222222222222222222222422222222121211224211222122412112
212121111212121112212211411211114112421111112111121114111
12112221111211111211114211211142111111111224111244421121
1122111111121112112141111121111242411221111122112121121121

#103 (mean 1.579, $|D_3|=0.229$):
2222222221211111211121112241122221121211122122111221111122
1221111212421121121212111212412121121222442121112121442211
2112121211111124221211212121124221111221111122211112121112
12141211121121212211222111241222124121114211111121241121112

#104 (mean 1.587, $|D_3|=0.217$):
121142221212121212111421112211112212211222141211212122221
1111214112111211211221141111122212111122111112111122122222
211421211211111211121411142141121222112211121421241221
21111121412221211412121122211211114221141122121221224211141

#105 (mean 1.592, $|D_3|=0.229$):
11111111111111111111111111111212122111112122112211211411221
1121211111111211221112212224211121421124212221121222422222
11242121211112122211224411122222421222112111222124221222241
22211224112112124111221212121242112221121121112121241122212

#106 (mean 1.583, $|D_3|=0.232$):
1111121142221221212111241111121122224114221122122111422111
21112222121112211222122211211211122221211121212121242221122
2121221111121122111122121112221114224211214221122112412142
122141114121212122211111211112242111221211112211122111221112

#107 (mean 1.579, $|D_3|=0.218$):
242222222221221111111122214121222111112211221111111111111
121211111114212112211111422224112121111211111111121212221
222121222111121112121142211112121211412142112111221212212211
1121111114412212211114222121412222111242221141222211111424

#108 (mean 1.579, $|D_3|=0.214$):
242222222222212122121122111212112212211211112224214414122
111211111121111411211112221111112214211112211111142211122
111111222222221222114141111414211121211111112112121411212
1112214144111222111121211121111124211222211111211211141

#109 (mean 1.592, $|D_3|=0.220$):
1111111111112242124214214112211121421222121122111242111214
21112121111122141141421211121242111111111221221111242142121
11214221222211112111112412422111111212121111212211211212
4121111112211112111211111121211142422211412242122221112122

#110 (mean 1.592, $|D_3|=0.220$):
1111111111222142112122212211111141121421112412411211142121
211111211111111124112121211112222221241121111422221221122
2422222111112114212111111121112112244211121111121212124211
11221221111211111214111221421142122122124211421214112111214

#111 (mean 1.579, $|D_3|=0.229$):
222222212122121121111212121121111112114212221221111141212
12411212122121111122121221124111112221221112112111211222111
1224141111111221124441212212121111121122421222112122112221
1121212211122112121121212411212112211121242111211221422421

#112 (mean 1.583, $|D_3|=0.218$):
111111111121211121141221221412211122211111211141111122111
24112142121111212121241142211242211112111222144141214222121
11212112122211122212111211211111112122111112112111121121121
21244211141112221212211222112221112442422111211144221211111

#113 (mean 1.579, $|D_3|=0.248$):
222242222222222222112122211122221112111142211121212111241
111112212121212242111121122211221121122121211121122111111
11222211221111212122211211114221142121211214141112211112
2124112112111121122112112214122221121422212121422211111112

#114 (mean 1.579, $|D_3|=0.229$):
2222222222222222211111211211121212124111122411112222
1122222122111121111211211141122211412411111222112411211
21121421211111111144121122212211121211421241121211221222
142212121122221212221121211212114111211211211112214211

#115 (mean 1.592, $|D_3|=0.220$):
11111111111111112112112111422221221222242212211211114111
222212121112111111114411142212121214111242214124211212112
12121141112122122211111241214422111222122121114121121241211
122211441121121211211121112111142111122241142221211111

#116 (mean 1.592, $|D_3|=0.233$):
111111111111111111111111111111111111112112112114212221244
112111422124111121241222121142222121242122112121222111222212
221414412114111111212124211412142141212421221111121112222
11211111112114121121112411111224222141121222124124111121111

#117 (mean 1.579, $|D_3|=0.222$):
222222211222211114111122214211221212422112211111122241221
2112112112112221112111212111211112211222122421111121211422
1111112111121222241111121211221111211121421111221121222
41222211221214111112141222221111212114124211241222114112221

#118 (mean 1.587, $|D_3|=0.214$):
12412224222141112121111121212112122111212111211121422111
2112211111211111221141111222111122212111211121111242212111
11122112421112211114112111212211112142112242211122221224144
12422122221111142121112111124241121222221111121121221142

#119 (mean 1.579, $|D_3|=0.218$):
2222422222222222222222222222242112112112141111111221142
1112121112111212122212211211122121112211211111121124111
1112111221114211111121111124142122422141112222121111122
1411211221111111211211412112211111121121111141422241211

#120 (mean 1.579, $|D_3|=0.214$):
22222422222422222212241121212211112111111121211211212111
21221112112111211111121141221122111211221112224141112111
12112212121122114121121211412211114111221211111211422444
4112111112121122241141221241241112121121211111121121211221